Dalton Cole (He/Him)
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Education

University of Missouri - Columbia — July 2021
Ph.D. Computer Science

Missouri University of Science and Technology — May 2019
M.S. Computer Science
B.S. Computer Science
Minor: Computer Engineering & Mathematics

Work Experience

Sandia National Laboratories — Albuquerque, NM
Senior Computer Scientist — July 2021 - Present

- Supported the development and analysis of wide-view visual-based satellite remote sensing sensors
- Designed a graph neural network architecture for event classification using PyTorch
- Lead a team of engineers working on analyzing remote sensing data
- Spearheaded a project converting image segmentation maps into realistic images using a generative adversarial neural network
- Proposed and received funding for machine learning and data science projects
- Mentored Ph.D. and undergraduate interns in software development

Cybersecurity Intern — May 2017 - Dec 2019

- Applied machine learning to find correlations between source code and binary
- Wrote emulytics software using C for the 1553 data bus protocol
- Created a Laika BOSS module to filter out metadata
- Designed and implemented a graph labeling attack algorithm
- Developed Python libraries in C++ for more efficient software
- Analyzed network data using Bro and machine learning

Missouri University of Science and Technology — Rolla, MO
Graduate Teaching Assistant — Fall 2019

- Instructed students on objected-oriented programming (C++)
- Prepared lesson plans that are both informative and engaging to students
- Enforced the high integrity standards outlined by the university

Cerner — Kansas City, MO
Software Engineering Intern — Summer 2016

- Created a Ruby On Rails web application with added security features
- Developed or used existing APIs for tools like Jenkins and Crucible
- Led a team of interns in weekly live demos to management
- Won crowd favorite at the intern hackathon

Additional Experience

Personal Website — July 2020 - Present

- Implemented a personal website using Django
- Managed a PostgreSQL database
- Extensively used Django templates to promote concise and rapid development
- Created a mobile-first responsive website design using bootstrap
- Containerized website deployment via Docker
- Used TravisCI and Github Actions to automate unit testing and deployment
- Incorporated Google Search for quick and efficient website queries

Programming Competitions — Fall 2015 - 2019

- Won MS&T's Programming Cup - Individual's Bracket
- ACM ICPC Regional's first place on-site

Publication

- **D Cole**, S Newman, D Lin. *"A New Facial Authentication Pitfall and Remedy in Web Services".* IEEE Transactions on Dependable and Secure Computing, vol. 19, no. 4, pp. 2635-2647, 1 July-Aug. 2022

Computer Skills

Advanced:	Python	C++	Rust					
Proficient:	Java	JavaScript	HTML	Linux	LaTeX	SQL	Django	
	PyTorch	Tensorflow	Keras	Docker	Matplotlib	Numpy	Sklearn	